OPTO

Operational manual for OpenVnmrJ

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1. INTRODUCTION:

This manual is included with the installer and can be found in the unzipped folder under the file name "`OPTO_OVJ_Manual.pdf`". The manual provides a comprehensive guide to all OPTO settings and capabilities, including an example run to demonstrate how OPTO can be utilized for OpenVnmrJ.

To download the latest version of OPTO for OpenVnmrJ, please go to https://git.doit.wisc.edu/nmrfam-public/opto-releases/-/blob/main/README-OVJ.md

2. INSTALLATION GUIDE:

Prerequisites

1) Opto requires Java version 8 or later. Please install a Java package like OpenJDK. Make sure you have not installed a “headless” version of Java. The “apt info” command can show you what version (if any) of Java you have installed and whether or not it is headless (as the “headless” term will be part of the installed package name).

2) Opto requires that OpenVNMRJ has already been installed. The OPTO installation program will install OPTO into the OpenVNMRJ application directory.

Process:

a) Download the most recent `opto-ovj.zip` file from the opto-releases website.
b) Launch a terminal.
c) If you have an older version of Opto installed execute sudo uninstall-opto.sh. The uninstall program should already be in the path.
d) Use the `cd <directory-path>` command to navigate to the directory that contains `the downloaded opto-ovj.zip`.
e) Execute the `unzip opto-ovj.zip` command to extract the contents of the `opto-ovj.zip` file.
f) Switch to the directory that was just extracted.
g) Install opto by running the `sudo ./install.sh` command.
h) If you wish, you can remove the files that were unzipped.

3. OPENING OPTO:

a) Within OpenVnmrJ command line, type ‘listenon’ to allow OpenVnmrJ to communicate with OPTO. If you have communications issues please add an OPTO_PORT environmental variable to your environment with a known open port.

b) After ‘listenon’ is activated open OPTO by typing opto.sh in the terminal command line.

4. INITIAL SETUP:

4.1 SELECTING THE REGION OF INTEREST FOR THE INTEGRATION:

In the OPTO GUI in the Job panel there is an Integration option shown. You choose between Time and Frequency. When shimming you usually choose “Time”. When optimizing parameters you usually use “Freq”. When the “Freq” option is selected, OPTO utilizes the integrated region using the normal region integration tool. To define and save a region of interest:

a) Acquire a 1D spectrum.

b) Set the integration to partial by typing intmod='partial'
c) Assign the region you wish to integrate by clicking integration button along sidebar.
d) Define the region of interest.

OPTO reads this info when integrating the signal region during operation. It may be changed at any time during operation,

Otherwise, clear the integrated region by typing ‘cz’ in the command line. This will integrate the full spectral width.

If multiple regions are stored, OPTO only uses the first region.

5. RUNNING OPTO FOR SHIMMING:

Shimming: Using OPTO you can create a job that optimizes shims. You select shims of interest and initial values and ranges before running.

Prerequisites: Have a valid shim job defined. Have a sample loaded and MA adjusted. Have acquired an initial spectra.

Defining a shim job:

Press the New Job button. A dialog will launch. Press the Load Shims button in the lower left corner. Save the job with a name of your choosing.

The shimming process:

a) Have inserted in the probe the shimming sample. Spin sample up to the desired spinning rate.
b) Zero out all shim values by putting “opto(zero)” in the Setup field in the OPTO GUI job panel. Select desired shims to optimize using checkboxes in all the “Pass N” dialog tabs.

c) Load previously used ^{1}H to ^{13}C cross polarization and decoupling parameters for the current spinning rate into the spectrometer. Ensure the decoupling power and acquisition time are safe.
d) Acquire a single spectrum of adamantane without changing shims using current set of shim parameters for comparison to OPTO.
e) Set the acquisition time to 50 ms, or alternatively, on the OPTO GUI set the **Before** command to issue to the spectrometer software the acquisition time to be 50ms.
f) In pass 1 select the first order shims (x^1, y^1, z^1).
g) In pass 2 select the second order shims (xy, xz, yz, z^2, x^2y^2).
h) In pass 3 select the third order shims (x^3, y^3, z^3, xz^2, yz^2, zxy, zx^2y^2).
i) In pass 4 select the highest order of shims (z^4, z^3x, z^3y, z^5, z^4x, z^4y).
j) For additional improvements to the shimming, run the above passes 1-4 twice more after they have converged with setting the acquisition time to 200 ms for the first replicate, and then set the acquisition time to 400 ms for the last replicate.
k) Use default number of iterations, range, and tolerance for experiment. Click **Run Job**.
l) Confirm that the spectrometer is performing the experiment and is pulsing. The spectrometer controlling software should be updating as the experiments are running live.
m) Once the Run is complete: If wanted, the .log files of the optimization are written out to the experiment relevant directory for *posthoc* processing and analysis.

6. RUNNING OPTO FOR CP:

Parameter Optimization: Using OPTO you can create jobs that optimize parameters. You create and select parameters of interest and initial values and ranges before running.

a) In OVJ create parameters of interest. For instance, with maclib files associated with your pulse sequence of interest.

b) In the OPTO GUI click the New Job button. One by one add your parameters of interest in the dialog and set any initial values and ranges as appropriate. Save the job with a name of your choosing.

c) **Sample Preparation:**
   a. Have inserted in the probe the Cross Polarization compatible sample. Spin sample up to the desired spinning rate. We use crystalline N-acetyl-Valine for setting CP conditions (between 2 kHz to 24 kHz ω_r) or microcrystalline GB1(25kHz +)

d) **Experiment Setup:**
   a. Open OPTO with opto.sh command.
   b. Load pulse sequence with ^{1}H to ^{13}C cross polarization option. (We use the hX pulse sequence with its associated parameters for this example). Ensure the parameter
   c. Load your defined CP job that is related to the pulse sequence.
   d. Check that decoupling parameters are safe for the acquisition time chosen.
   e. Estimate the power levels wanted for the Hartmann-Hahn condition for the sample in the spectrometer. Depending on the spinning rate chosen, these can be

double quantum or single quantum conditions.

e) **Initial Spectrum Acquisition:**
   a. Here the user is expected to have a theoretical starting point get an initial CP signal. A previously optimized CP experiment can be a good starting point here.
   b. Acquire a 1D cross polarization spectrum for comparison.
   c. Once the unoptimized 1D is finished and before running OPTO job, ensure that you are in the correct experimental page in the OPTO GUI. Press **Resync** otherwise and check parameters.

f) **Select Region of Interest:**
   a. Select the region of interest. Please refer to section 4.1 above.

g) **Modify Parameters:**
   a. Edit the GUI panels to manage the parameters you want to optimize throughout each pass.

      Note: Ensure that all parameter names used are the same naming convention as in the maclib files.

h) **Set Cross Polarization optimization job:**
   a. In the OPTO GUI in the Job panek select the 'Freq' integration method for best CP optimization results. Ensure the correct region of spectrum is selected.
   b. In pass 1, select the parameters that control the power levels of the ^{1}H and ^{13}C channels.
   c. In pass 2, select the parameters that control the power levels of the ^{1}H and ^{13}C channels and the ramp power delta.
   d. In pass 3, select the parameters that control the power levels of the ^{1}H and ^{13}C channels, the ramp power delta, and the ramp shape.
   e. Check that the spectrometer software has the appropriate number of scans, dummy scans, acquisition time, and integration range selected for operation. If not, these parameters may be written into the **Setup** tab to be issued to the spectrometer before operation.
   f. Once the parameters and commands are set and the operator is happy with how they expect the Job to be performed, one may save the entirety of the job using the **Save Job** button and then can load the optimization protocol to a different experiment with the **Load Job** button.

i) **Run Optimization:**
   a. Use the default settings for the tolerance and range. Change them if needed.
   b. Click on '`Run job`'.
   c. The results are written out to the terminal opto.sh was launched from as the run progresses.

j) **Completion and Application:**
   a. Once the Run is complete: If wanted, the .log files of the optimization are written out to the experiment relevant directory for *posthoc* processing and analysis.

7.Citation:

All publications resulting from the use of this software should cite the following paper:

OPTO: Automated Optimization for Solid-State NMR Spectroscopy, Collin G. Borcik, Barry DeZonia, Thirupathi Ravula, Benjamin D. Harding, Rajat Garg, and Chad M. Rienstra *Journal of the American Chemical Society* **2025** *147* (4), 3293-3303 DOI: 10.1021/jacs.4c13295